Assignment 1 – Task 01

Object Oriented Programming (C++) | Resize Array

Required function

The program creates an array with a size entered by the user, then expands it to twice its original size. The first half contains the original values, and the second half contains twice those values.

Function prototype

```
int* resizeArray(int inArray[], const int size);
```

Complete C++ solution

```
#include <iostream>
using namespace std;

// Create a new array with twice the original size.
int* resizeArray(int inArray[], const int size)
{
    int* newArray = new int[size * 2];

    // Copy the original elements.
    for (int i = 0; i < size; i++)
    {
        newArray[i] = inArray[i];
    }

    // Store twice each original value in the second half.
    for (int i = 0; i < size; i++)
    {
        newArray[size + i] = inArray[i] * 2;
    }

    return newArray;
}

int main()
{
    int size;

    cout << "Enter size of the array: ";
    cin >> size;

    if (size <= 0)
    {
        cout << "Invalid array size!" << endl;
        return 0;
    }

    int* arr = new int[size];

    cout << "Enter elements of array: ";
    for (int i = 0; i < size; i++)
    {
        cin >> arr[i];
    }

    int* resizedArray = resizeArray(arr, size);

    cout << "Resized array: ";
    for (int i = 0; i < size * 2; i++)
    {
        cout << resizedArray[i] << " ";
    }
    cout << endl;

    // Release dynamically allocated memory.
```

```
    delete[] arr;
    delete[] resizedArray;

    return 0;
}
```

Sample execution

```
Enter size of the array: 4
Enter elements of array: 1 2 3 4
Resized array: 1 2 3 4 2 4 6 8
```

How it works

1. **new int[size * 2]** allocates a new array with double the capacity.
2. The first loop copies all original elements into the new array.
3. The second loop places each original element multiplied by 2 in the second half.
4. The function returns a pointer to the new array.
5. **delete[]** releases both dynamically allocated arrays and prevents memory leaks.

Note: The original array is not modified. The returned array is a separate dynamically allocated array.